IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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https://github.com/Preeta-DE/SpaceX-IBM-Capstone

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Executive Summary

- Built an end-to-end SpaceX Falcon 9 landing prediction pipeline using API data, web scraping, data wrangling, SQL, visualization, interactive analytics, and machine learning.
- Created a binary landing outcome label where successful landings were encoded as 1 and unsuccessful outcomes as 0.
- EDA showed that launch site, orbit type, payload mass, and operational maturity influence landing success patterns.
- SQL analysis highlighted four unique launch sites, 45,596 kg NASA CRS payload mass, and the first successful ground landing on 2015-12-22.
- Interactive analytics were delivered through Folium maps and a Plotly Dash dashboard with site filters, pie charts, and payload-based scatter plots.
- Classification models reached 83.33% test accuracy, with Decision Tree achieving the highest cross-validation score during tuning.

Introduction

SpaceX advertises Falcon 9 launches at about $62M, while competitor launches can exceed $165M; first-stage reusability is a major reason for this cost advantage. By determining if the first stage will land, we can determine the price of the launch. This project harnesses machine learning to predict first-stage reusability, empowering more efficient launch cost estimations.

Business question: can we predict whether the Falcon 9 first stage will land successfully using historical launch data?

Project objective: collect launch data, prepare it for analysis, discover patterns, build interactive analytics, and train classification models for landing prediction.

Main analytical questions:

Which sites and orbit types have higher success rates?

How do payload and flight history relate to outcomes?

Which model best predicts landing success?

Section 1

Methodology

Methodology

- Collected Rocket Launch data through the SpaceX REST API and Falcon 9 Data from Wikipedia using web scraping.
- Wrangled the dataset by identifying orbit frequencies, successful vs. failed landings, and labelling missions accordingly.
- Performed EDA using seaborn/matplotlib visualizations and SQLite SQL queries to identify operational patterns.
- Built interactive analytics using Folium maps and a Plotly Dash dashboard for site and payload exploration.
- Prepared features, standardized data, tuned multiple classification models with GridSearchCV, and compared model performance

Data Collection

- Two complementary collection methods were used: SpaceX API calls for structured launch metadata and Wikipedia scraping for historical launch records.
- The SpaceX REST API workflow extracted:
- The web scraping workflow parsed the Falcon 9 / Falcon Heavy historical launch information into a pandas dataframe for downstream analysis.
- All the obtained data columns by respective techniques are as follows :

SpaceX REST API	Web Scraping
FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, Latitude	**Flight No., Launch site, Payload, PayloadMass, Orbit, Customer, Launch outcome, Version Booster, Booster landing, Date, Time**

- GitHub Link: https://github.com/Preeta-DE/SpaceX-IBM-Capstone

Data Collection - SpaceX API

- Requested past launches from the SpaceX REST API endpoint and normalized the JSON response into a dataframe.
- Used helper functions to retrieve booster version, payload mass, orbit, launch site, coordinates, landing outcome, flights, gridfins, reuse status, legs, landing pad, block, reuse count, and serial.
- This API-driven dataset provided the structured foundation for wrangling, EDA, geospatial analysis, and machine learning.
- GitHub Link: https://github.com/Preeta-DE/SpaceX-IBM-Capstone/blob/main/01_SpaceX-Data-collection-api.ipynb

API workflow

1. Sent a GET request to the SpaceX launch API to retrieve historical mission data.
2. Converted the JSON response into a Python dictionary using .json() and flattened it with json_normalize() into a Pandas dataframe.
3. Called additional SpaceX API endpoints via custom helper functions to enrich launches with rocket, payload, core, and launchpad details.
4. Combined all launch information into a single consolidated dataframe ready for downstream analysis.
5. Filtered the dataset to keep only Falcon 9 launches, matching the capstone business focus.
6. Imputed missing values in the Payload Mass column using the mean to maintain dataset completeness.
7. Saved the cleaned dataset as a CSV file for subsequent EDA, SQL analysis, visualization, and predictive modelling.

Data Collection - Scraping

- Used requests and BeautifulSoup to scrape Falcon 9 and Falcon Heavy launch records from a static Wikipedia snapshot.
- Parsed launch date, booster version, launch site, payload, payload mass, orbit, customer, launch outcome, and booster landing fields.
- Converted the HTML table into a pandas dataframe and exported it for downstream use.
- GitHub Link: https://github.com/Preeta-DE/SpaceX-IBM-Capstone/blob/main/02_SpaceX-Webscraping.ipynb

Web scraping workflow

1. Sent a GET request to the Wikipedia page containing Falcon 9 launch data.
2. Parsed the HTML with BeautifulSoup to represent the page structure programmatically.
3. Identified and extracted all column headers from the HTML table holding the launch records.
4. Iterated through table rows and cells to collect detailed launch information.
5. Organized the collected data into a Python dictionary using the headers as keys.
6. Converted this dictionary of launch details into a Pandas dataframe.
7. Saved the resulting dataframe as a CSV file for subsequent analysis and integration with the main project dataset.

Data Wrangling

- Cleaned launch records and inspected launch site counts, orbit counts, and landing outcomes.
- Defined unsuccessful outcomes as {False ASDS, False Ocean, False RTLS, None ASDS, None None}.
- Created the classification target column Class, where 1 = successful landing and 0 = unsuccessful landing.
- Prepared the cleaned dataset for EDA, SQL analysis, dashboarding, and predictive modeling.
- GitHub Link: https://github.com/Preeta-DE/SpaceX-IBM-Capstone/blob/main/03_SpaceX-Data wrangling.ipynb

EDA with Data Visualization

- Used seaborn and matplotlib to explore relationships between launch outcome and operational variables.
- Plotted Flight Number vs Launch Site, Payload vs Launch Site, Success rate by Orbit, Flight Number vs Orbit, Payload vs Orbit, and yearly success trend.
- These visualizations helped identify patterns in experience, site behavior, orbit performance, and payload-related success.
- GitHub Link: https://github.com/Preeta-DE/SpaceX-IBM-Capstone/blob/main/05_EDA_Data_Visualization.ipynb

1. **Flight Number vs. Launch Site - Scatter plot:** Identify patterns between the order of launches (Flight Number) and the locations from which they took off (Launch Site).
2. **Payload vs. Launch Site - Scatter plot**: Explore relationships between the weight of the cargo (Payload) and the launch location (Launch Site).
3. **Success Rate by Orbit Type - Bar chart**: Visualize the success rate of missions for each orbit type, showing what percentage of launches succeeded in each category.
4. **Flight Number vs. Orbit Type - Scatter plot:** Uncover patterns between the launch order (Flight Number) and the orbit targeted by each mission (Orbit Type).
5. **Payload vs. Orbit Type - Scatter plot:** Explore relationships between the cargo weight (Payload) and the type of orbit targeted by the launch (Orbit Type).
6. **Launch Success Yearly Trend - Line plot:** Show yearly trends in launch success rates to see whether success is improving, declining, or staying consistent over time.

EDA with SQL

- Loaded launch data into SQLite to answer business-oriented analytical questions using SQL.
- Queried unique launch sites, NASA payload totals, average payload by booster version, first successful ground landing date, mission outcomes, and other filtered launch records.
- SQL results were used to summarize operational evidence in a concise, presentation-friendly format.
- GitHub Link: https://github.com/Preeta-DE/SpaceX-IBM-Capstone/blob/main/04_EDA_SQL_sqllite.ipynb

Queries:

- Names of unique launch sites
- First 5 records where the launch site begins with “CCA”
- Total payload mass carried by boosters launched for NASA (CRS) missions
- Average payload mass carried by booster version F9 v1.1
- Date of the first successful landing on a ground pad
- Names of boosters that successfully landed on a drone ship with payload mass between 4,000 and 6,000 kg
- Total number of successful and failed missions
- Names of booster versions that have carried the maximum payload
- Failed landing outcomes on a drone ship, with their booster version and launch site, for months in the year 2015
- Count of landing outcomes between 2010-06-04 and 2017-03-20, ordered in descending count

Interactive Map with Folium

- Marked and labeled all launch sites on a map using latitude and longitude coordinates.
- Added color-labeled success and failure markers for each launch and explored proximity to infrastructure such as coastline, railway, and highway.
- This geospatial analysis supported the insight that launch-site location and nearby features matter operationally.
- GitHub Link: [https://github.com/Preeta-DE/SpaceX-IBM-Capstone/blob/main/06_SpaceX_Visual Analytics_Folium.ipynb](https://github.com/Preeta-DE/SpaceX-IBM-Capstone/blob/main/06_SpaceX_Visual_Analytics_Folium.ipynb)

- **Launch sites**: Circular markers show all launch site locations, with text labels such as "NASA Johnson Space Center." These markers highlight where the sites are on the map and how close they are to the equator and nearby coastlines.
- **Launch outcomes**: Green markers indicate successful launches and red markers indicate failed launches. Marker clusters group nearby points so it is easy to spot sites with many successful or unsuccessful missions.
- **Distance to nearby features**: Lines connect a selected launch site (for example, CCAFS SLC-40) to nearby railways, highways, the coastline, and the closest city. These lines show how close each launch site is to key infrastructure and population areas.

Dashboard with Plotly Dash

- Developed an interactive dashboard with a launch-site dropdown, a success pie chart, a payload range slider, and a payload-success scatter plot.
- Users can compare all sites or isolate a single site to explore success patterns by payload and booster category.
- The dashboard complements static EDA by enabling interactive filtering and visual exploration.
- GitHub Link: https://github.com/Preeta-DE/SpaceX-IBM-Capstone/blob/main/07_SpaceX_Visual_Analytics_Plotly_Dash.py

1. **Dropdown list**
   Lets users filter the dashboard by launch site. When a site is selected, all charts focus on launches from that location.
2. **Pie chart**
   Shows the total number of successful launches across all sites. When a specific site is chosen, it updates to display the success and failure rates for that site only.
3. **Slider**
   Allows users to choose a range of payload mass values. This helps analyse how launch success relates to different payload weights.
4. **Scatter chart**
   Plots payload mass against launch success, coloured by booster version category. Combined with the dropdown and slider, it reveals how payload and booster type relate to outcomes at each launch site

Predictive Analysis (Classification)

- Prepared features using one-hot encoding and standardized numeric inputs before model training.
- Split data into training and test sets, then used GridSearchCV to tune Logistic Regression, SVM, Decision Tree, and KNN models.
- Evaluated models with accuracy and confusion matrices to determine the strongest landing predictor.
- Identified the best model using Jaccard_Score, F1_Score and Accuracy.
- GitHub Link: https://github.com/Preeta-DE/SpaceX-IBM-Capstone/blob/main/08_SpaceX_PredictiveAnalytics.ipynb

1. **Data Preparation**
   - Selected relevant features, including the Class target column.
   - Standardized the feature data using StandardScaler to improve model performance.
2. **Train/Test Split**
   - Split the dataset into training and testing sets using train_test_split for fair model training and evaluation.
3. **Model Selection & Tuning:**
   - Used GridSearchCV with 10-fold cross-validation to tune hyperparameters.
   - Applied this process to Logistic Regression, SVM, Decision Tree, and KNN models.
4. **Model Evaluation**
   - Evaluated each tuned model on the test set using accuracy via the .score() method.
   - Inspected confusion matrices to understand prediction errors and correct classifications.
5. **Model Selection**
   - Compared models using Jaccard score and F1 score in addition to accuracy.
   - Selected the model with the strongest overall performance across these three metrics.

Results

Exploratory Data Analysis:

- Launch success has improved over time
- KSC LC-39A has the highest success rate among landing sites
- Orbits ES-L1, GEO, HEO and SSO have a 100% success rate

Visual Analytics:

- Most launch sites are near the equator, and all are close to the coastline.
- Launch sites are far away enough from anything a failed launch can damage (city, highway, railway), while still close enough to bring people and material to support launch activities.

Predictive Analytics:

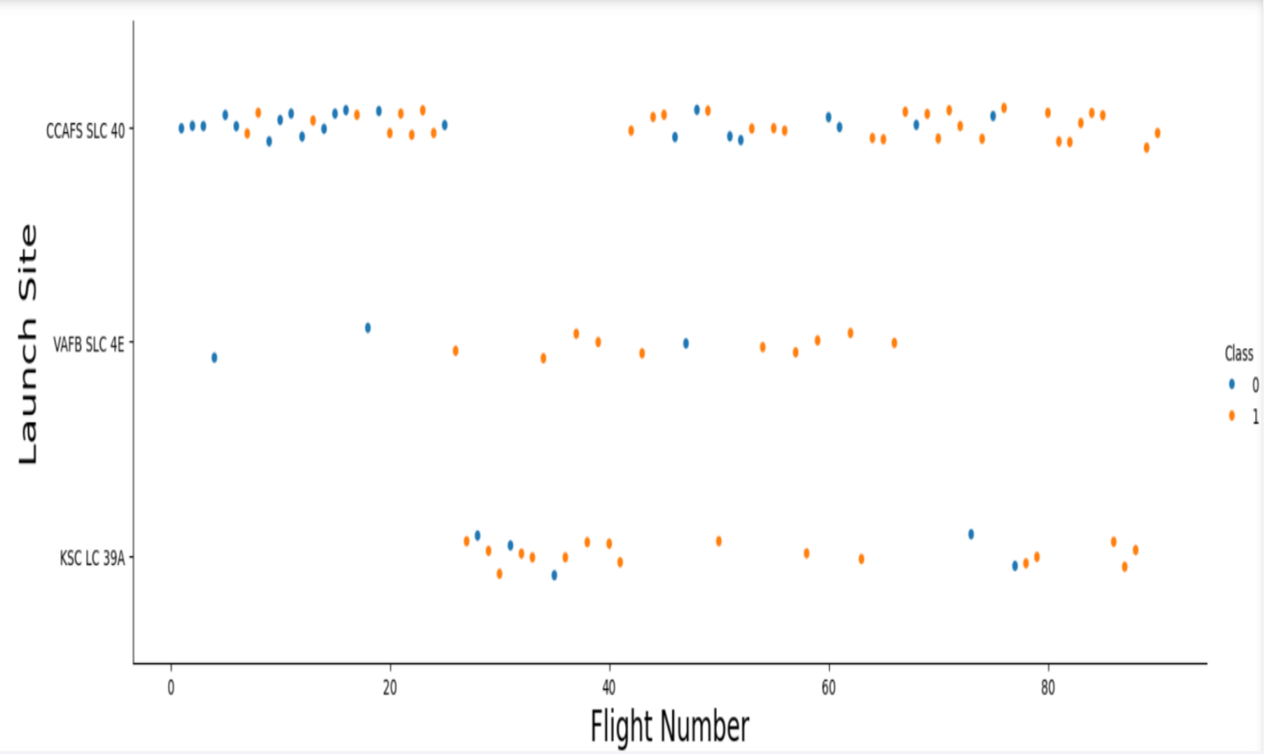
- Decision Tree model is the best predictive model for the dataset

Section 2

Insights drawn from EDA

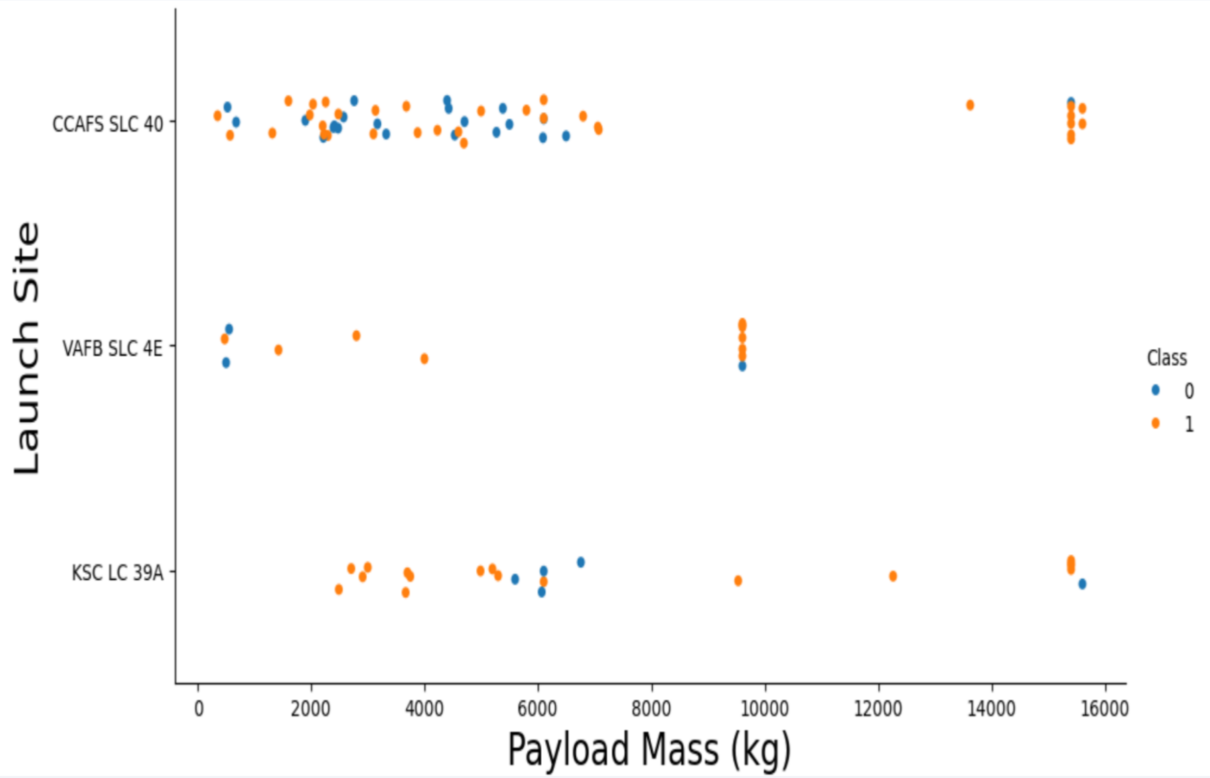
Flight Number vs. Launch Site

- Earlier flights had a lower success rate (blue = fail)
- Later flights had a higher success rate (orange = success)
- Around half of launches were from CCAFS SLC 40 launch site
- VAFB SLC-4E and KSC LC-39A have higher success rates
- We can infer that new launches have a higher success rate
- Insight : Mission experience improved first-stage landing reliability



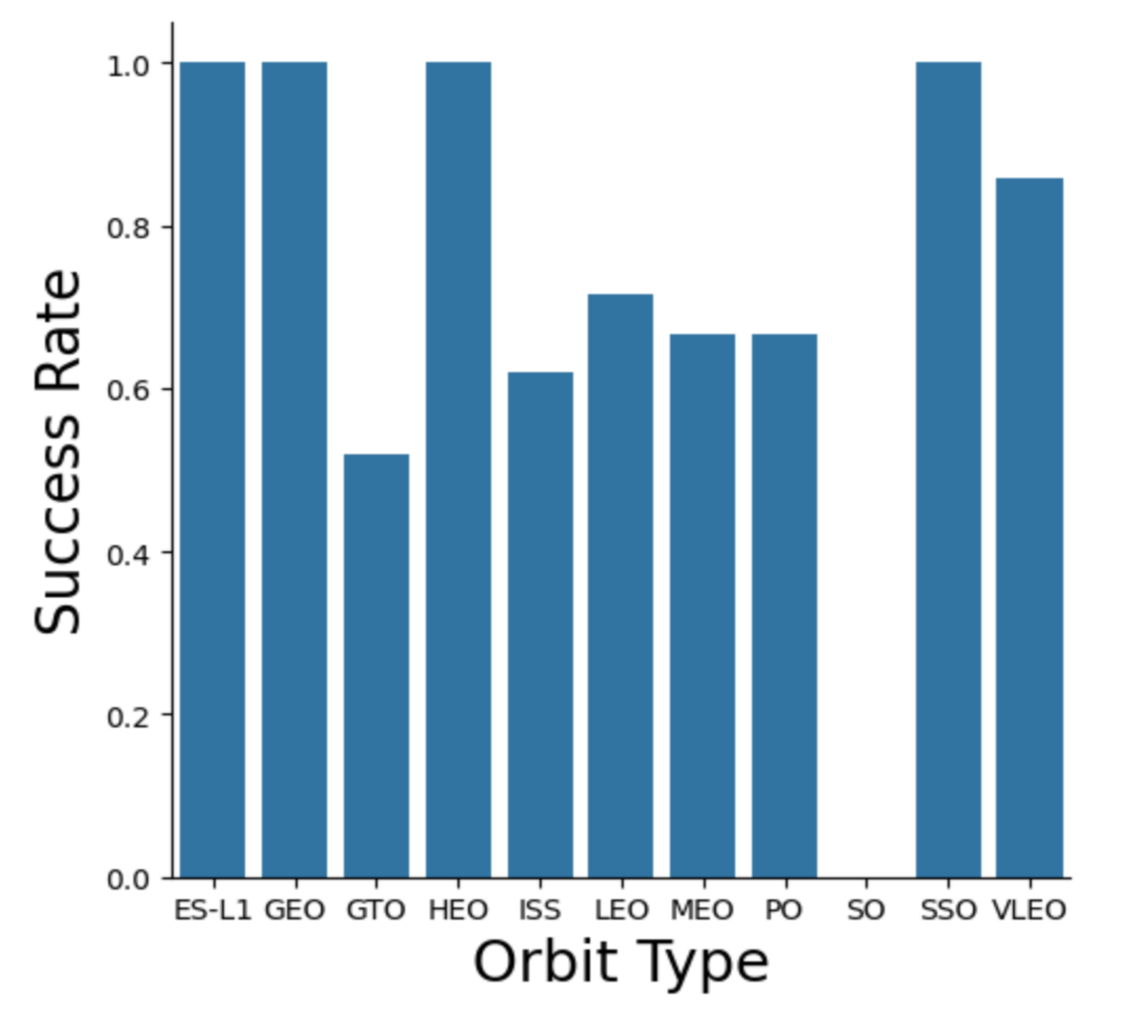
Payload vs. Launch Site

- Typically, the higher the payload mass (kg), the higher the success rate
- Most launches with a payload greater than 7,000 kg were successful
- KSC LC-39A has a 100% success rate for launches less than 5,500 kg
- VAFB SLC-4E has not launched anything greater than ~10,000 kg
- Insight: Higher payload mass has higher success rate



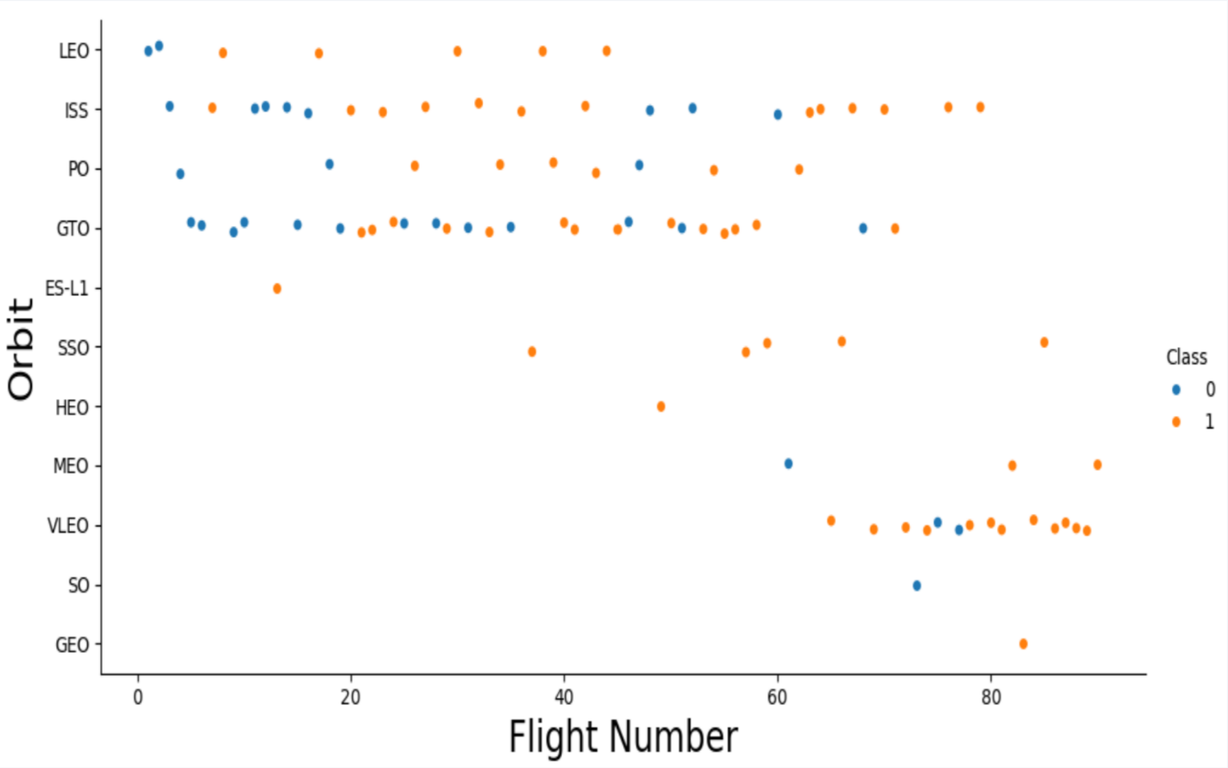
Success Rate vs. Orbit Type

- ES-L1, GEO, HEO and SSO orbit categories show consistently stronger success rates than others, indicating mission profile matters.
- Insight: Orbit should be treated as an important predictive feature.



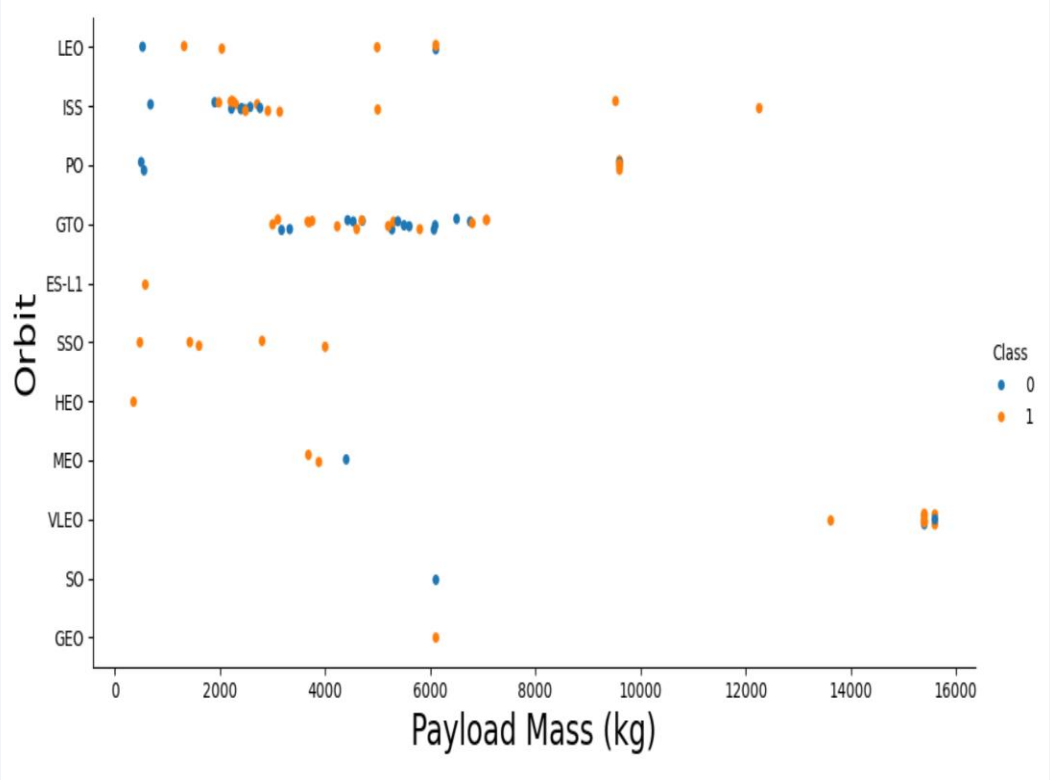
Flight Number vs. Orbit Type

- The success rate generally increases as the number of flights grows for each orbit.
- This pattern is especially strong for the LEO orbit. However, GTO orbit does not follow this trend.
- Insight: Time and experience interact with orbit complexity.



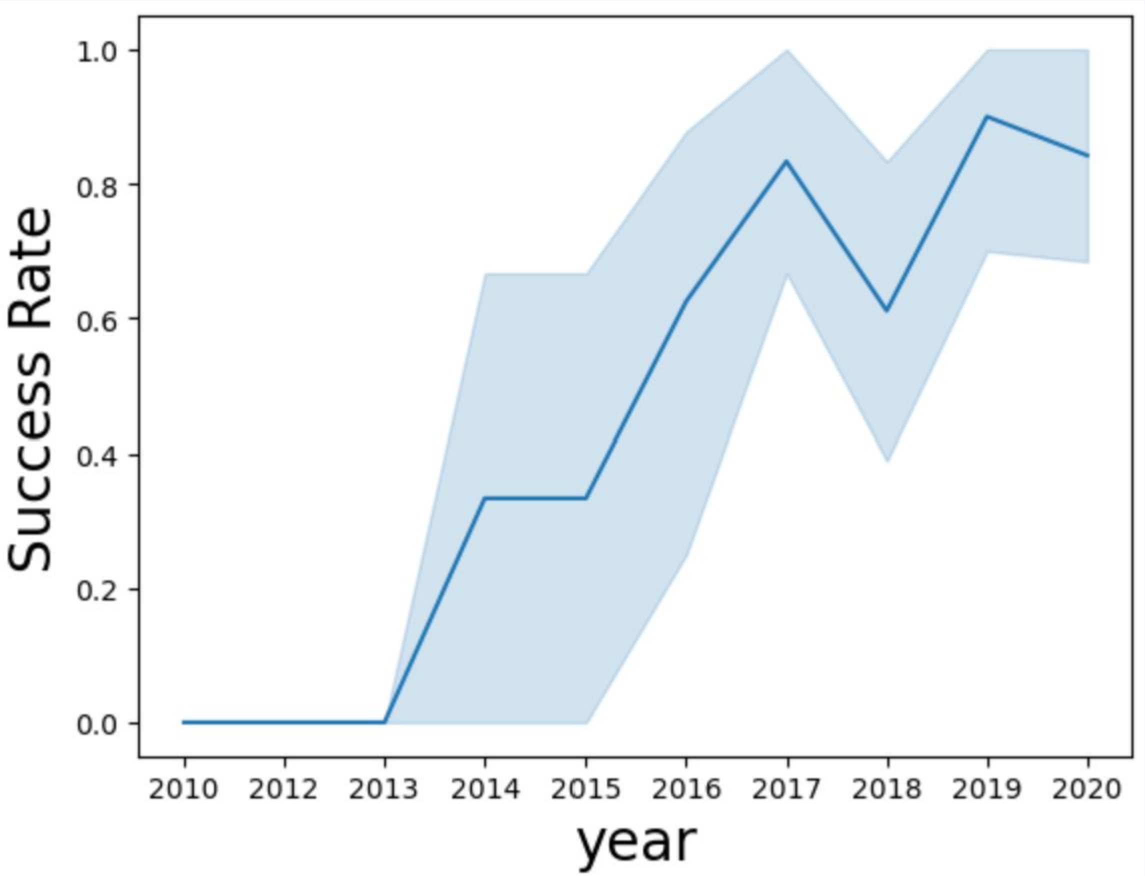
Payload vs. Orbit Type

- Heavy payloads are better with LEO, ISS and PO orbits.
- The GTO orbit has mixed success with heavier payloads.
- Insight: Payload and Orbit jointly influence the likelihood of successful recovery.



Launch Success Yearly Trend

- Success rate improved from 2013-2017 and 2018-2019.
- The yearly average success rate trends upward over time.
- This supports the interpretation that SpaceX improved landing capability through reuse, design iteration, and operational experience.
- Insight: Later missions have higher landing success than early ones.



All Launch Site Names

- Unique launch sites identified in the SQL analysis: CCAFS LC-40, CCAFS SLC-40, VAFB SLC-4E, and KSC LC-39A.

Display the names of the unique launch sites in the space mission

```
%sql SELECT DISTINCT(Launch_Site) FROM SPACEXTABLE
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Displays 5 records of Launch Site Names beginning with ‘CCA’.

Display 5 records where launch sites begin with the string 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site like "%CCA%" LIMIT 5
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Total payload carried by boosters for NASA CRS missions: 45,596 kg.

Display the total payload mass carried by boosters launched by NASA (CRS) ¶

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE CUSTOMER = 'NASA (CRS)'
```

```
 * sqlite:///my_data1.db
Done.
```

SUM(PAYLOAD_MASS__KG_)
45596

Average Payload Mass by F9 v1.1

- Average of 2928.4 kg was carried by booster version F9 v1.1

Display average payload mass carried by booster version F9 v1.1

```
%sql Select AVG(PAYLOAD_MASS__KG_) from SPACEXTABLE WHERE Booster_Version = 'F9 v1.1'
```

```
 * sqlite:///my_data1.db
Done.
```

AVG(PAYLOAD_MASS__KG_)
2928.4

First Successful Ground Landing Date

- The first successful landing outcome on a ground pad occurred on 2015-12-22.

```
%sql SELECT min(DATE) FROM SPACEXTABLE WHERE Landing_Outcome like 'Success%ground pad%'
```

```
 * sqlite:///my_data1.db
Done.
```

min(DATE)
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- Boosters listed by SQL for successful drone ship landings with payload mass between 4,000 and 6,000 kg included F9 FT B1022, F9 FT B1026, F9 FT B1021.2, and F9 FT B1031.2.
- These records show that successful drone ship recovery was achieved even for relatively heavy payload missions.

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000 ¶

```
T Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome like 'Success (drone ship)%' and PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- SQL mission outcome counts show 98 Success Mission Outcome, with only 1 Failure (in flight) and 1 Success(payload status unclear).
- This confirms the high overall mission reliability of the Falcon program in the analyzed records.

List the total number of successful and failure mission outcomes

```
%sql SELECT Mission_Outcome, Count(Mission_Outcome) FROM SPACEXTABLE GROUP BY Mission_Outcome
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	Count(Mission_Outcome)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- SQL identified several Block 5 boosters among the boosters carrying the maximum payloads.
- Examples include B1048.4, B1049.4, B1051.3, and later reuse variants.

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
%sql SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ IN (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE)
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- In 2015, SQL identified failed drone-ship landing outcomes including boosters F9 v1.1 B1012 and F9 v1.1 B1015 from CCAFS LC-40

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
ome, Booster_Version, Launch_Site from SPACEXTABLE WHERE Landing_Outcome like "Failure (drone ship)%" and substr(Date,0,5)="2015"
```

```
 * sqlite:///my_data1.db
Done.
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Between the date 2010-06-04 and 2017-03-20, the Failure (drone ship) count of Landing Outcome is 5 whereas Success (ground pad) is 3.

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
utcome like "Success (ground pad)" and Date between "2010-06-04" and "2017-03-20" group by Landing_Outcome order by Count(*) desc
```

```
 * sqlite:///my_data1.db
Done.
```

Landing_Outcome	Landing_Outcome_Count
Failure (drone ship)	5
Success (ground pad)	3

Section 3

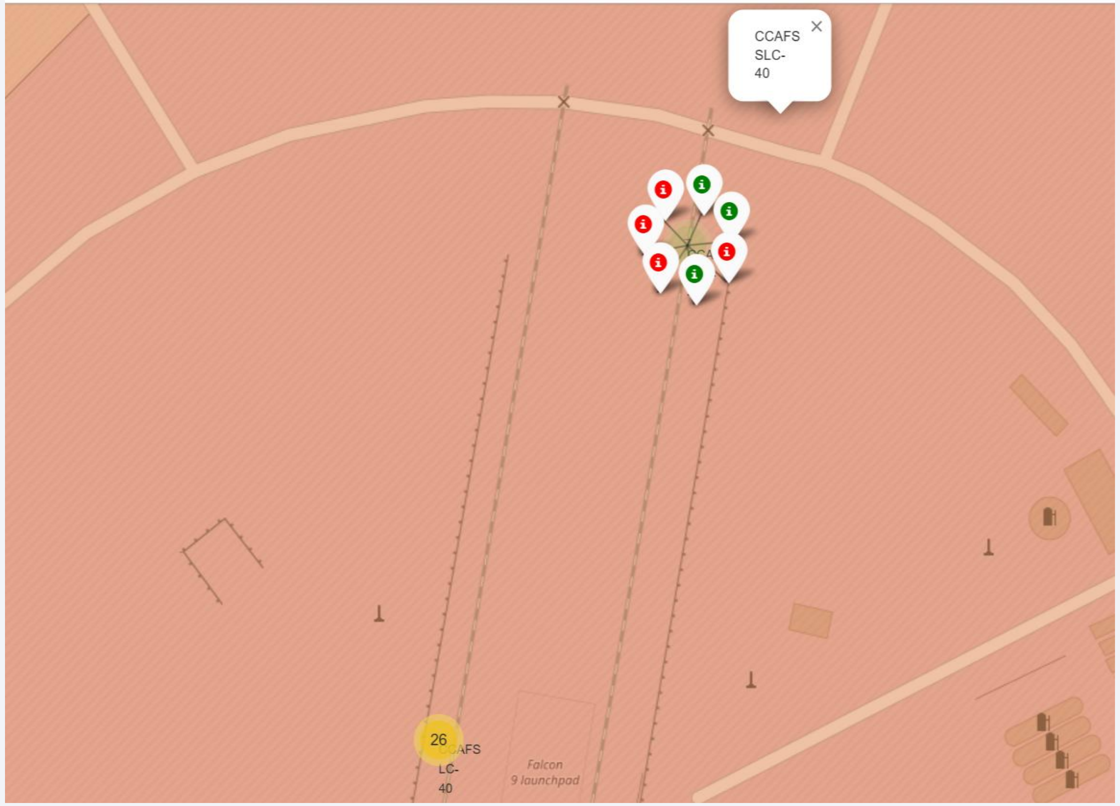
Launch Sites Proximities Analysis

Launch Sites with Folium Markers

- All major launch sites are positioned near the coast.
- This reduces public safety risk, since rockets carry large amounts of fuel and any failure is less likely to affect cities or residential areas.
- Equator proximity gives a natural boost to the rockets due to the rotational speed of earth.

Launch Outcomes

- At each launch site, green markers indicate successful launches and red markers indicate unsuccessful launches.
- At CCAFS SLC-40, the displayed sample highlights a 3/7 success rate, or 42.9%.
- Colour-labelled markers in marker clusters identifies the launch sites with high success rates.



Distance to Proximities

CCAFS SLC-40 launch site:

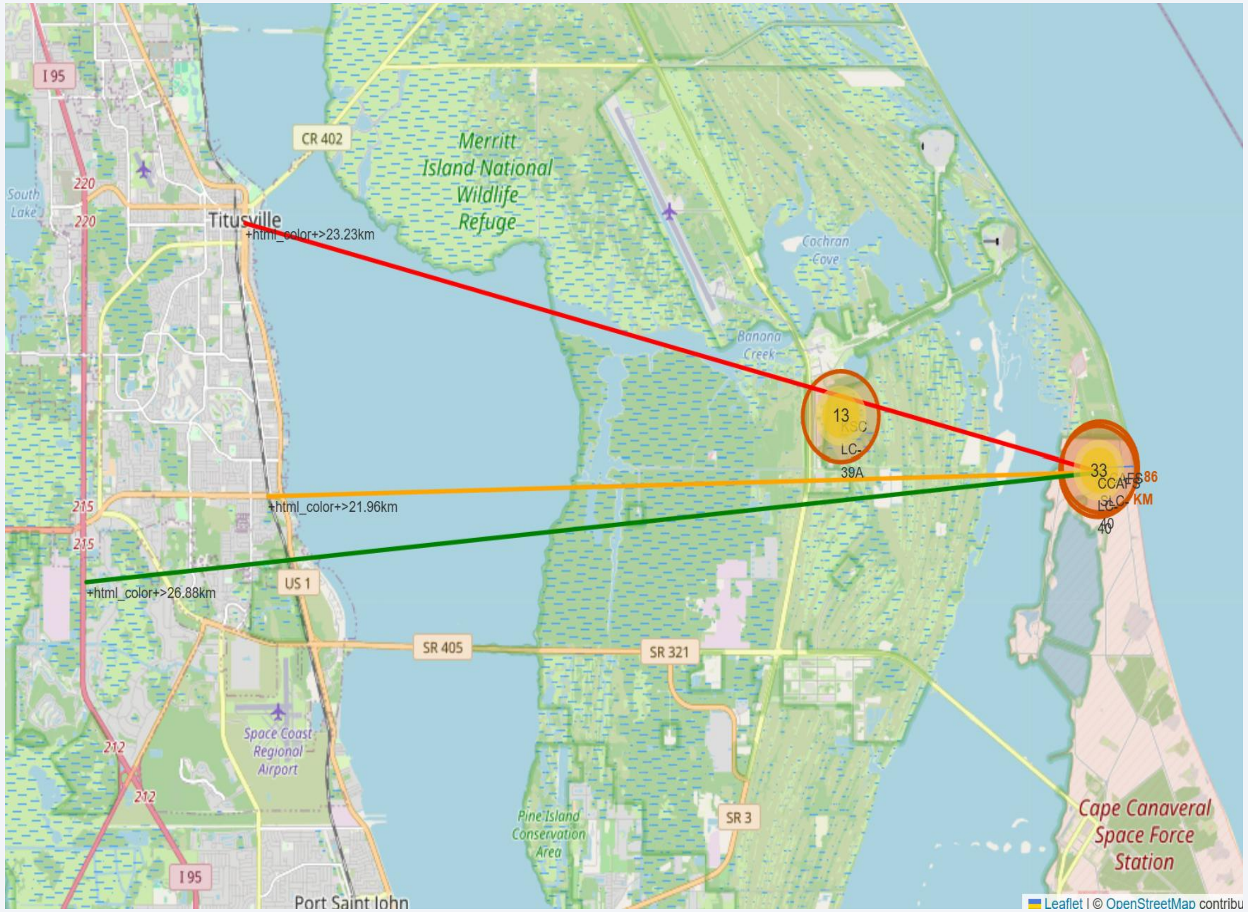
0.86 km distance from the nearest coastline.

21.96 km distance from the nearest railway.

23.23 km distance from the nearest city.

26.88 km distance from the nearest highway.

These features matter because launch sites require both safety buffers and logistical accessibility

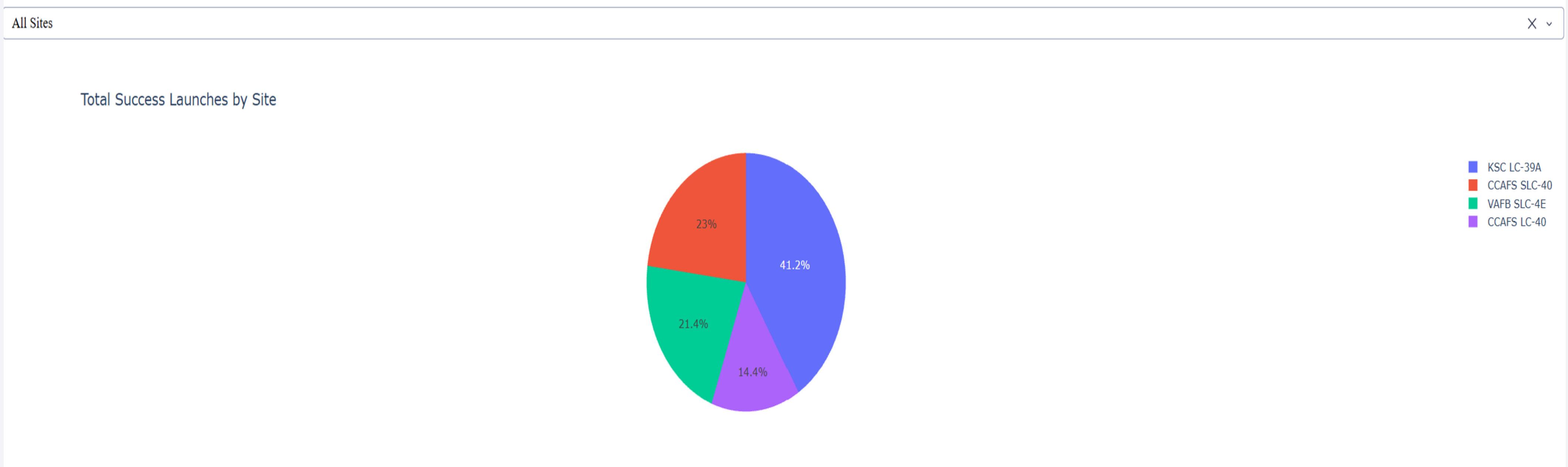


Section 4

Build a Dashboard with Plotly Dash

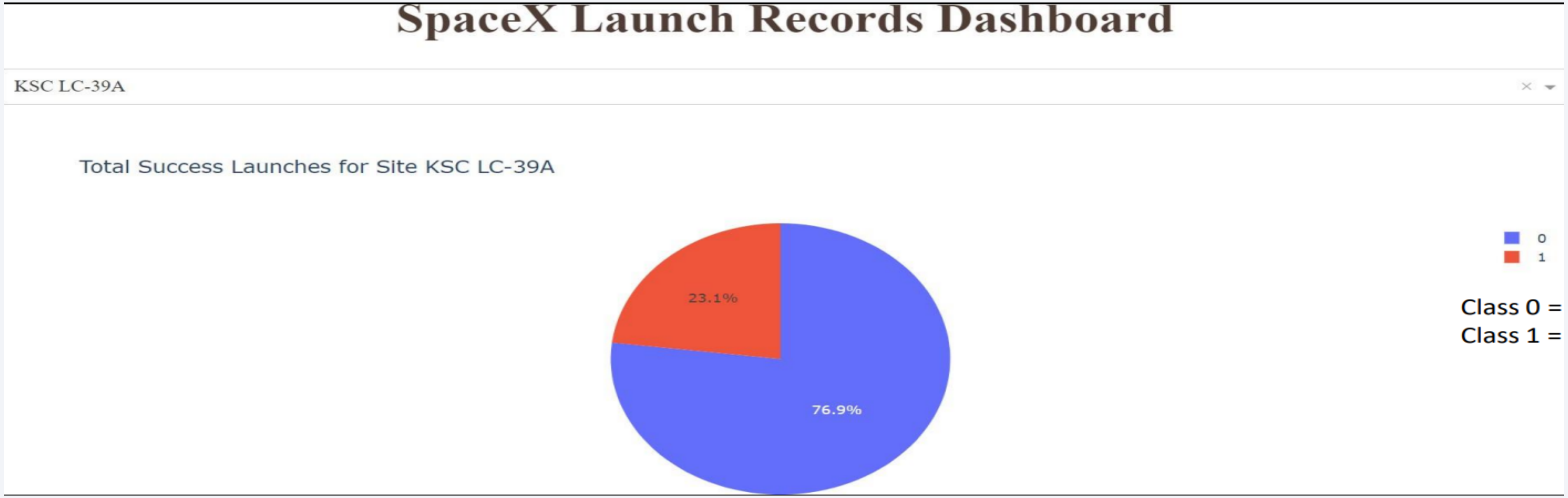
Launch Success by Site

- The Plotly Dash pie chart summarizes launch success across all sites using a site-selection dropdown.
- KSC LC-39A has the most successful launches amongst launch sites (41.2%) as opposed to CCAFS LC-40 with lowest successful launches of 14.4%



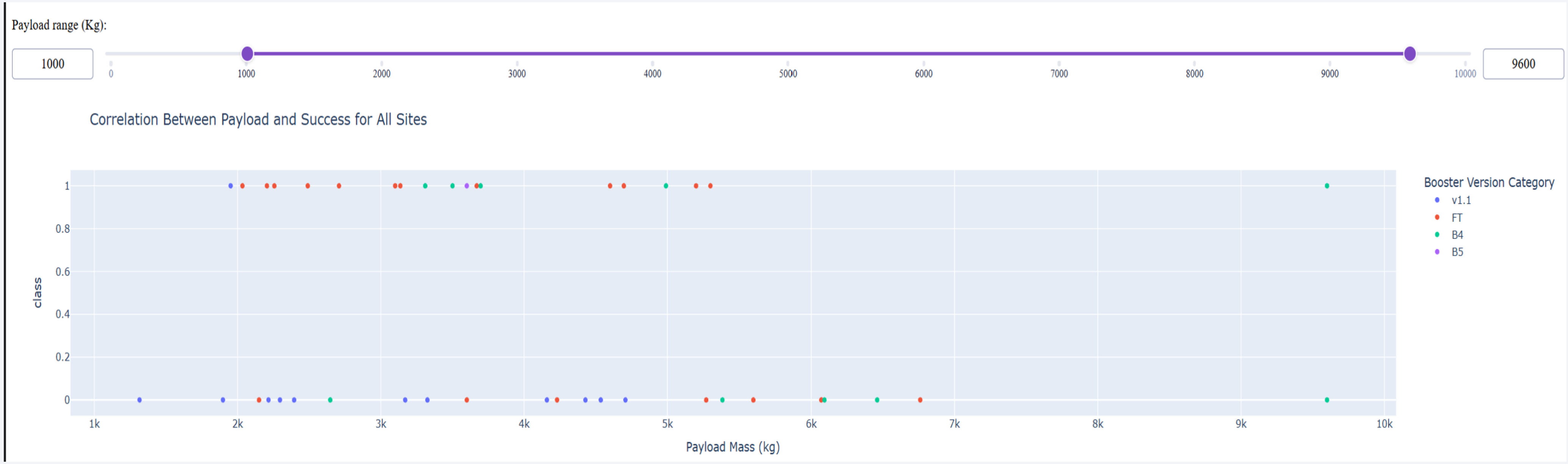
Launch Success (KSC LC-39A)

- KSC LC-39A has the highest success rate amongst launch sites - 76.9%
- 10 successful launches and 3 failed launches.



Payload Mass and Success

- Payloads between 2,000 kg and 5,000 kg have the highest success rate.
- Class 1 indicates success and 0 indicates failure.



Section 5

Predictive Analysis (Classification)

Classification Accuracy

- Test accuracy for Logistic Regression, SVM, Decision Tree, and KNN was 83.33% in the notebook outputs.
- During tuning, the Decision Tree reached the highest cross-validation score at 0.875, while Logistic Regression, SVM, and KNN were close behind.
- Decision Tree model slightly outperformed the rest

Model selection note: all four models tied on test accuracy, but Decision Tree was chosen because it achieved the best cross-validation score during tuning.

	LogReg	SVM	Tree	KNN
Jaccard_Score	0.800000	0.800000	0.800000	0.800000
F1_Score	0.888889	0.888889	0.888889	0.888889
Accuracy	0.833333	0.833333	0.833333	0.833333

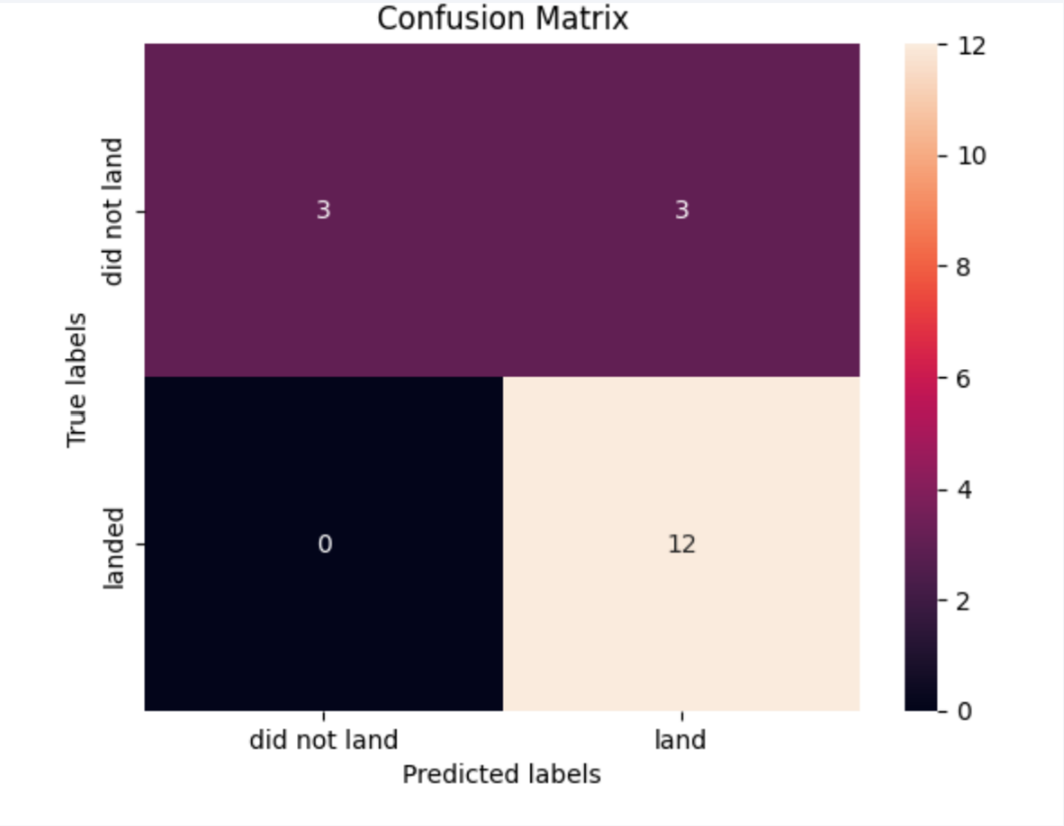
```
models = {'KNeighbors':knn_cv.best_score_,
              'DecisionTree':tree_cv.best_score_,
              'LogisticRegression':logreg_cv.best_score_,
              'SupportVector': svm_cv.best_score_}

bestalgorithm = max(models, key=models.get)
print('Best model is', bestalgorithm,'with a score of', models[bestalgorithm])
if bestalgorithm == 'DecisionTree':
    print('Best params is :', tree_cv.best_params_)
if bestalgorithm == 'KNeighbors':
    print('Best params is :', knn_cv.best_params_)
if bestalgorithm == 'LogisticRegression':
    print('Best params is :', logreg_cv.best_params_)
if bestalgorithm == 'SupportVector':
    print('Best params is :', svm_cv.best_params_)
```

```
Best model is DecisionTree with a score of 0.875
Best params is : {'criterion': 'entropy', 'max_depth': 2, 'max_features': 'sqrt', 'min_samples_leaf': 1,
```

Confusion Matrix

- The confusion matrix shows how well the classifier predicted landing outcomes on the test set.
- Correct predictions: 12 successful landings and 3 failed landings.
- Errors: 3 failed launches were incorrectly predicted as successful landings; no actual successful landings were missed.
- Overall accuracy: 83.3%



Conclusions

Key takeaways

- Launch success has improved steadily over time, with recent missions showing strong reliability and higher overall success rates.
- Launch sites are strategically located near the equator and coasts, leveraging Earth's rotation for efficiency while prioritizing safety over populated areas.
- KSC LC-39A stands out as the top-performing site, leading in successful launches overall and achieving perfect success for lighter (<5,500 kg) payloads.
- More complex orbits show varied success, while ES-L1, GEO, HEO, and SSO orbits currently achieve 100% success, indicating strong capability for high-value missions.
- Heavier payloads are increasingly launched successfully, suggesting growing confidence in reusability and vehicle performance at higher mass ranges.
- Across models, performance was similar on the test set, with the Decision Tree model slightly outperforming alternatives and serving as a practical choice for predicting launch success.
- Business implication: stronger landing prediction improves launch-cost estimation and helps explain SpaceX's reusable-launch advantage.

Appendix

- GitHub Link: https://github.com/Preeta-DE/SpaceX-IBM-Capstone

Thank you!